

A Adjusted EF definition

$$EF = \frac{|G_T \cap H|}{(N_L + N_{\tilde{G}} + N_{\text{miss}}) h_{\text{rand}}}$$

$$h_{\text{rand}} = |H|/|L|$$

- $\mathcal{G}$ all valid genes
- L genes measured in this screen
- H hits in the screen library
- G_T genes acquired over T rounds

B Effective-budget terms

$$N_L = |G_T \cap L|$$

acquired genes measured in the screen

$$N_{\tilde{G}} = |G_T \setminus \mathcal{G}|$$

hallucinated acquisitions

$$N_{\text{miss}}$$

unfilled acquisition slots

$$n_{\text{eff}} = N_L + N_{\tilde{G}} + N_{\text{miss}}$$

total effective budget used

C Worked comparison: raw hits versus enrichment

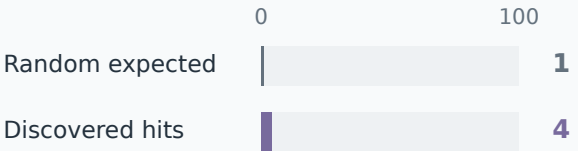
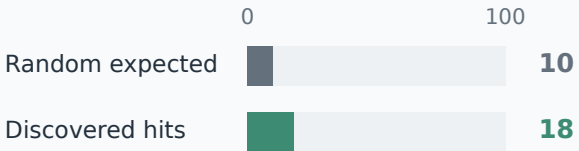
Screen A

- **Hit-Rich:** $h_{\text{rand}} = 10\%$
- **In a batch of 100 genes,** random expects **10 hits.**

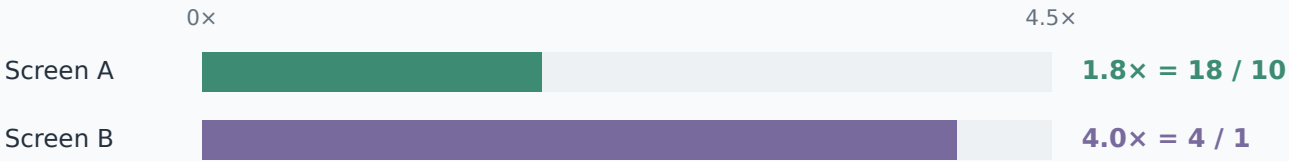
Screen B

- **Hit-Sparse:** $h_{\text{rand}} = 1\%$
- **In a batch of 100 genes,** random expects **1 hit.**

Raw hits in a 100-gene batch

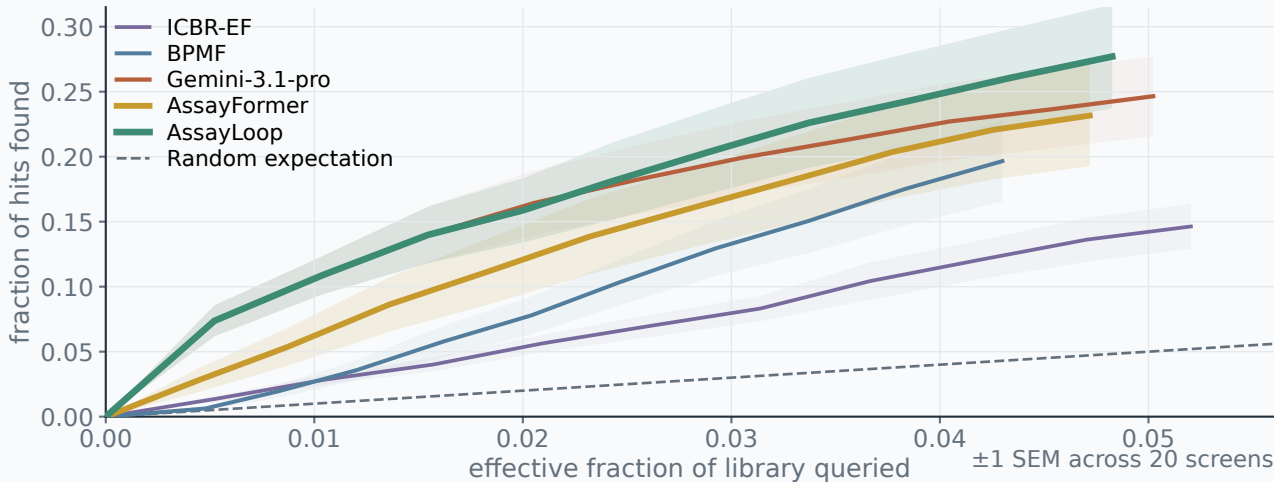


Enrichment factor

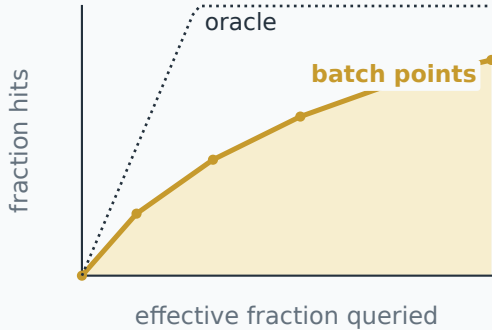


More raw hits can still represent less enrichment over random.

D Actual recovery trajectories



E Adjusted nAUC definition



$$x_t = \frac{\sum_{i=0}^t (N_{L,i} + N_{\tilde{G},i})}{|L|} \quad y_t = \frac{\sum_{i=0}^t N_{H,i}}{|H|} \quad \text{nAUC} = \frac{\text{AUC}}{\text{AUC}_{\text{oracle}}}$$

F Complementary diagnostics

- Fraction of hits**
 $|G_T \cap H|/|H|$
endpoint yield
- Shortfall**
 $|G_T \setminus L|/|G_T|$
unscoreable proposals
- % essential**
 $|G_T \cap H \cap E|/|G_T \cap H|$
universal vs specific hits
- Vendi ratio**
effective diversity / batch
higher = more distinct
- Pathway overlap**
pairwise Jaccard / random
>1 = more redundant
- Effective pathways**
 $EP = \exp(H(p))$
higher = broader coverage